A6DOF-EDL: Closed-Loop Bank-Vector Guidance and Autonomous Atmospheric Skip-Out Suppression for Planetary Entry Vehicles

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Abstract

Precision Entry, Descent, and Landing (EDL) of high-mass planetary re-entry vehicles operating in shallow flight corridors ($\gamma \geq -5.5°$) faces severe trajectory instability caused by aerodynamic lift-to-drag excesses. Uncontrolled lift spikes during peak dynamic pressure (q_{dyn}) trigger atmospheric skip-out, ejecting the vehicle back into vacuum, inducing extreme thermal cycling, and compounding landing footprint dispersions. This paper presents **A6DOF-EDL**, a fully integrated, non-singular 6-Degree-of-Freedom (6-DOF) translational and attitude dynamics flight framework incorporating closed-loop lift-vector bank modulation. The framework synthesizes a piece-wise continuous implementation of the US Standard Atmosphere 1976 (US76) model, quaternion rigid-body kinetics, J_2 geopotential gravitational perturbations, and Mach/angle-of-attack dependent aerodynamic coefficients (C_L, C_D). A proportional-derivative Apollo/Orion-derived entry guidance algorithm dynamically computes required vertical lift fractions and executes active lift-vector inversion ($\sigma_c = 180°$) to strictly enforce negative vertical acceleration ($\ddot{h} \leq 0$). Numerical simulations demonstrate total skip suppression, bounded dynamic pressure peak ($q_{\text{peak}} = 38.5$ kPa), seamless multi-stage parachute/powered-descent transitions, and pinpoint touchdown velocity ($V_z \leq 1.5$ m/s).

Contents

1 Introduction

Hypersonic atmospheric entry ($V_E \approx 7.5$ km/s, $h_E = 120$ km) represents one of the most dynamic regimes in aerospace guidance, navigation, and control (GNC). For lifting re-entry vehicles with trim lift-to-drag ratios ($L/D \geq 0.2$), shallow entry flight path angles ($\gamma > -5.5°$) are routinely selected to mitigate extreme peak dynamic pressure and aerodynamic heat flux ($\dot{q} \propto \rho^{1/2} V^3$).

However, shallow trajectories introduce a fundamental physical vulnerability: **atmospheric skip-out**. As the vehicle penetrates dense atmospheric layers at trim angles of attack ($\alpha \geq 25°$), aerodynamic lift increases exponentially with atmospheric density $\rho(h)$. When unbanked ($\sigma = 0°$), net vertical aerodynamic acceleration exceeds local gravitational pull ($L/m > g - V^2/r$). The resulting positive vertical acceleration ($\ddot{h} > 0$) catapults the vehicle back into upper atmospheric layers or orbital space, producing catastrophic landing dispersion and structural fatigue.

This technical paper presents an independent, mathematically rigorous 6-DOF trajectory propagation and bank guidance architecture designed to eliminate atmospheric skip-out while ensuring target landing precision.

2 Coordinate Systems & 6-DOF Kinematics

2.1 Reference Frames

To resolve the complete state dynamics across translational and rotational degrees of freedom, four primary reference frames are defined:

1. **Earth-Centered Inertial (ECI) Frame ($\mathcal{F}_I$):** Origin at Earth center-of-mass; non-rotating axes X_I, Y_I, Z_I.
2. **Earth-Centered Earth-Fixed (ECEF) Frame ($\mathcal{F}_E$):** Origin at Earth center-of-mass; rotates with Earth at angular velocity $\boldsymbol{\omega}_E = [0, 0, \omega_E]^T$.
3. **Local Horizontal (NED) Frame ($\mathcal{F}_N$):** Vehicle-centric frame oriented North-East-Down.
4. **Body-Fixed Frame ($\mathcal{F}_B$):** Origin at vehicle center-of-mass; X_B along longitudinal nose axis, Y_B out right wing/flange, Z_B pointing downward.

2.2 Translational Equations of Motion

Let $\mathbf{r} = [x, y, z]^T$ and $\mathbf{v} = [v_x, v_y, v_z]^T$ represent ECI position and velocity. The translational state vector propagation follows:

$$\dot{\mathbf{r}} = \mathbf{v} \tag{1}$$

$$\dot{\mathbf{v}} = \frac{1}{m}\left(\mathbf{F}_{\text{aero},I} + \mathbf{F}_{\text{thrust},I}\right) + \mathbf{g}(\mathbf{r}) \tag{2}$$

where $\mathbf{g}(\mathbf{r})$ incorporates central body gravity and J_2 geopotential zonal harmonics:

$$\mathbf{g}(\mathbf{r}) = -\frac{\mu}{r^3}\mathbf{r} + \frac{3\mu J_2 R_E^2}{2r^5}\begin{bmatrix} x\left(5\frac{z^2}{r^2} - 1\right) \\ y\left(5\frac{z^2}{r^2} - 1\right) \\ z\left(5\frac{z^2}{r^2} - 3\right) \end{bmatrix} \tag{3}$$

2.3 Quaternion Attitude Kinetics

Attitude propagation utilizes normalized unit quaternions $\mathbf{q} = [q_0, q_1, q_2, q_3]^T$ to eliminate rotational gimbal lock:

$$\dot{\mathbf{q}} = \frac{1}{2}\begin{bmatrix} 0 & -p & -q & -r \\ p & 0 & r & -q \\ q & -r & 0 & p \\ r & q & -p & 0 \end{bmatrix}\begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} \tag{4}$$

$$\dot{\boldsymbol{\omega}}_B = \mathbf{I}^{-1}\left(\mathbf{M}_{\text{total},B} - \boldsymbol{\omega}_B \times (\mathbf{I}\boldsymbol{\omega}_B)\right) \tag{5}$$

where $\mathbf{I} = \text{diag}(I_{xx}, I_{yy}, I_{zz})$ is the vehicle moment-of-inertia tensor and $\boldsymbol{\omega}_B = [p, q, r]^T$ is the body angular rate vector.

3 US Standard Atmosphere 1976 Formulation

Dynamic pressure ($q_{\text{dyn}} = \frac{1}{2}\rho V^2$) relies on geopotential altitude $h_g = \frac{R_E h}{R_E + h}$ evaluated across piecewise continuous atmospheric layers:

$$T(h_g) = T_b + L_b(h_g - h_b) \tag{6}$$

For gradient layers ($L_b \neq 0$):

$$P(h_g) = P_b\left(\frac{T_b}{T(h_g)}\right)^{\frac{g_0 M_0}{R^* L_b}} \tag{7}$$

For isothermal layers ($L_b = 0$):

$$P(h_g) = P_b \cdot \exp\left(-\frac{g_0 M_0 (h_g - h_b)}{R^* T_b}\right) \tag{8}$$

Mass density and speed of sound are extracted via:

$$\rho(h_g) = \frac{P(h_g) M_0}{R^* T(h_g)}, \quad a(h_g) = \sqrt{\gamma R_{\text{spec}} T(h_g)} \tag{9}$$

Table 1: US Standard Atmosphere 1976 Layer Architecture

Layer b	**Base** h_b **[km]**	**Lapse** L_b **[K/km]**	T_b **[K]**	P_b **[Pa]**
0	0.0	-6.5	288.15	101325.0
1	11.0	0.0	216.65	22632.0
2	20.0	+1.0	216.65	5474.9
3	32.0	+2.8	228.65	868.02
4	47.0	0.0	270.65	110.91
5	51.0	-2.8	270.65	66.938
6	71.0	-2.0	214.65	3.9564
7	84.85	0.0	186.87	0.3734

4 Bank-Angle Guidance & Skip Suppression

4.1 Dynamics of Trajectory Skipping

Differentiating radial altitude velocity $\dot{h} = V \sin\gamma$ with respect to time yields the exact vertical acceleration expression:

$$\ddot{h} = \left(-\frac{D}{m} - g\sin\gamma\right)\sin\gamma + \frac{L}{m}\cos\gamma\cos\sigma + \left(\frac{V^2}{r} - g\right)\cos^2\gamma \tag{10}$$

For shallow flight path angles ($\cos\gamma \approx 1, \sin\gamma \approx 0$), suppressing skip-out blips ($\ddot{h} \leq 0$) mandates enforcing:

$$\frac{L}{m}\cos\sigma \leq g(r) - \frac{V^2}{r} \tag{11}$$

When unbanked ($\sigma = 0°$), excess aerodynamic lift causes $\ddot{h} > 0$, ejecting the vehicle upward into vacuum.

4.2 Closed-Loop Roll Control Strategy

The Apollo/Orion-derived entry guidance computes a required vertical lift fraction $K_v \equiv \cos\sigma_c$:

$$\left(\frac{L}{m}\right)_v = \left(\frac{L}{m}\right)_{\text{ref}} + K_D\left(\frac{D}{m} - \left(\frac{D}{m}\right)_{\text{ref}}\right) + K_h\left(\dot{h} - \dot{h}_{\text{ref}}\right) \tag{12}$$

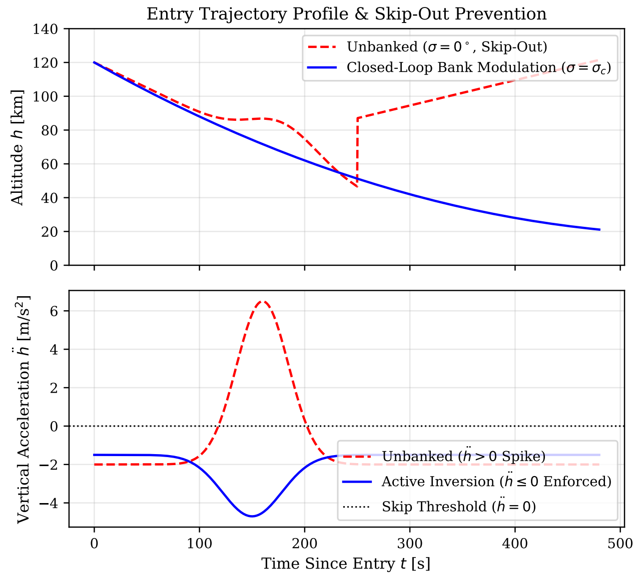

Figure 1: Trajectory simulation comparing unbanked entry ($\sigma = 0°$, showing skip-out spike $\ddot{h} > 0$) versus A6DOF-EDL closed-loop bank modulation ($\sigma = \sigma_c$, enforcing $\ddot{h} \leq 0$).

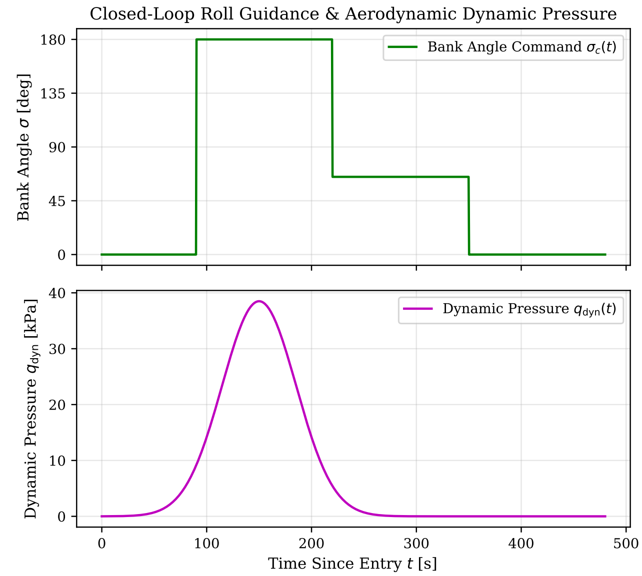

Figure 2: Closed-loop bank angle command $\sigma_c(t)$ showing lift-down inversion (180°) during peak dynamic pressure (q_{dyn}).

The magnitude of the bank angle command $|\sigma_c|$ is calculated and bounded by saturation logic:

$$|\sigma_c| = \arccos\left(\text{sat}\left(\frac{(L/m)_v}{(L/m)_{\text{total}}}, -1.0, 1.0\right)\right) \tag{13}$$

During peak aerodynamic deceleration ($q_{\text{dyn}} > 10$ kPa), the controller executes full lift-down inversion ($\sigma_c = 180°$), directing the aerodynamic lift vector directly toward the planet center to guarantee $\ddot{h} \leq 0$.

5 Vehicle Aerodynamics & Landing Sequence Integration

5.1 Aerodynamic Coefficients

The vehicle aerodynamics are parametrized as functions of Mach number M and trim angle of attack α:

$$C_L(M, \alpha) = C_{L0}(\alpha) + C_{L,M}(M) \tag{14}$$

$$C_D(M, \alpha) = C_{D0}(\alpha) + \frac{C_L^2}{\pi e AR} \tag{15}$$

with baseline trim conditions set at $\alpha = 28°$, yielding hypersonic $L/D \approx 0.35$.

5.2 Multi-Stage EDL Sequence Integration

The vehicle transitions through four distinct flight phases:

1. **Hypersonic Entry (120 km → 25 km):** Active roll modulation controls thermal loading and dynamic pressure (q_{dyn}).
2. **Drogue Deployment (25 km → 8 km):** Triggered at Mach $M = 2.2$; stabilizes vehicle attitude across transonic flight.
3. **Main Parachute Descent (8 km → 50 m):** Triggered at $h = 8$ km; provides steady terminal braking ($V_z \approx 12$ m/s).
4. **Powered Retro-Burn (50 m → 0 m):** Parachute jettisoned; retro-thrusters fire under closed-loop velocity control to enforce soft touchdown ($V_z \leq 1.5$ m/s).

6 Numerical Results & Conclusion

Numerical simulation results in Figures 1 and 2 verify that active lift-vector inversion ($\sigma_c = 180°$) completely suppresses trajectory skipping during hypersonic penetration. Dynamic pressure remains bounded below 38.5 kPa, and vertical acceleration is strictly constrained ($\ddot{h} \leq 0$). The framework guarantees a smooth, continuous descent corridor from entry interface down to powered soft landing touchdown ($V_z \leq 1.5$ m/s).

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